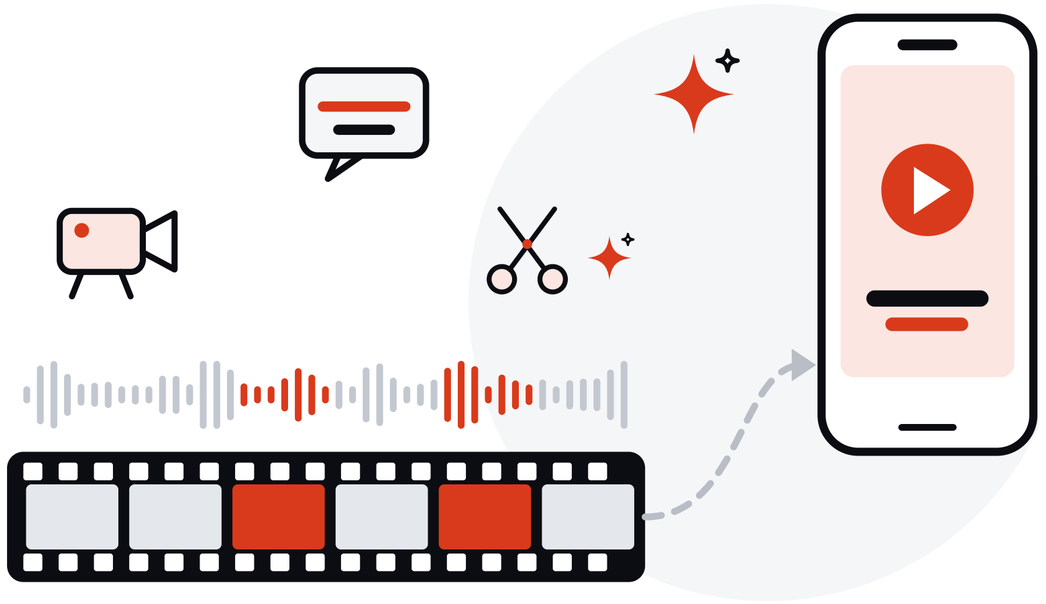
REC · MILESTONE 3

[SynopStudio]
MS3

Video editing and summarization with artificial intelligence

PEI — PROJETO EM ENGENHARIA INFORMÁTICA
· [2026/2027]



GROUP [15]

[Daniel Duque]	[124880]
[David Monteiro]	[125793]
[Lucas Reis]	[126287]
[Murilo Frezzato]	[125487]
[Tomas Lopes]	[125596]

SUPERVISORS

[Diogo Gomes]
[Rui Raposo]

deti
universidade de aveiro
departamento de eletrónica, telecomunicações e informática

Long talks turned into sharp clips that respect your voice.

GDPR

WHAT WE PROCESS



Account data
Email and Display name of creators



Voice and image
People who speak or appear in the recordings



Transcripts
Names, opinions and context in what was said



Voice profiles
Scoped to one video; would be biometric data (Art. 9) only if used to identify someone

ART. 6
Lawful basis
Contract; creators confirm permission at upload

ART. 9
Special categories
Voice profiles per project, never cross-matched

ART. 5
Minimisation and retention
Display name; videos deleted after 30 days

ART. 15–20
Data subject rights
Deleting a project or account removes everything

ART. 25 AND 32
By design and secure
TLS 1.3, AES-256, Keycloak

ART. 35
Impact assessment
DPIA required before launch

Licensing and intellectual property

Component	License	Usage
React, FastAPI, faster-whisper, pyannote.audio, Traefik	MIT	Library / separate service
Keycloak, SeaweedFS	Apache-2.0	Separate service
PostgreSQL	PostgreSQL License	Separate service
RabbitMQ	MPL-2.0	Separate service
FFmpeg	LGPL-2.1+	External process
yt-dlp	Unlicense	Library
pypdf, python-pptx, python-docx	BSD-3 / MIT	Library
Qwen3.8-27B	Apache-2.0	Model weights
SynopStudio	Apache-2.0	

Sector-specific regulations

Artificial Intelligence

RULE:
EU AI Act

WHAT IT MEANS:
People must know when content was made by AI

FOR US:
AI titles and descriptions are labelled, and nothing is exported without the creator's approval.

Audiovisual media and copyright

RULE:
Copyright and image rights

WHAT IT MEANS:
You can only use videos you own or are allowed to use, and people's image is protected.

FOR US:
The creator confirms authorship and licence before any processing.

Political and institutional communication

RULE:
Political advertising and elections

WHAT IT MEANS:
A clip must not change what someone said, and paid political ads have rules.

FOR US:
Every clip links to its original moment, and quotes are never invented.

Terms and conditions

01

Acceptable Use Policy

Minimum age 13. Mandatory license and consent confirmation; strict ban on out-of-context cuts, disinformation and hiding AI labels

02

Liability Disclaimers and AI

Provided “as is”. No liability for AI transcription or speaker errors; mandatory human review and approval before export

03

Refund/Cancellation Policies

Currently free; deleting projects/accounts removes all media. Future token plans allow 14 day proportional refunds.

04

Legal Jurisdiction

Governed by Portuguese and EU law, protecting consumer rights in their country of residence

05

Changes to the Terms

Reviewed annually. Material changes require 2 months’ advance notice via email and web app

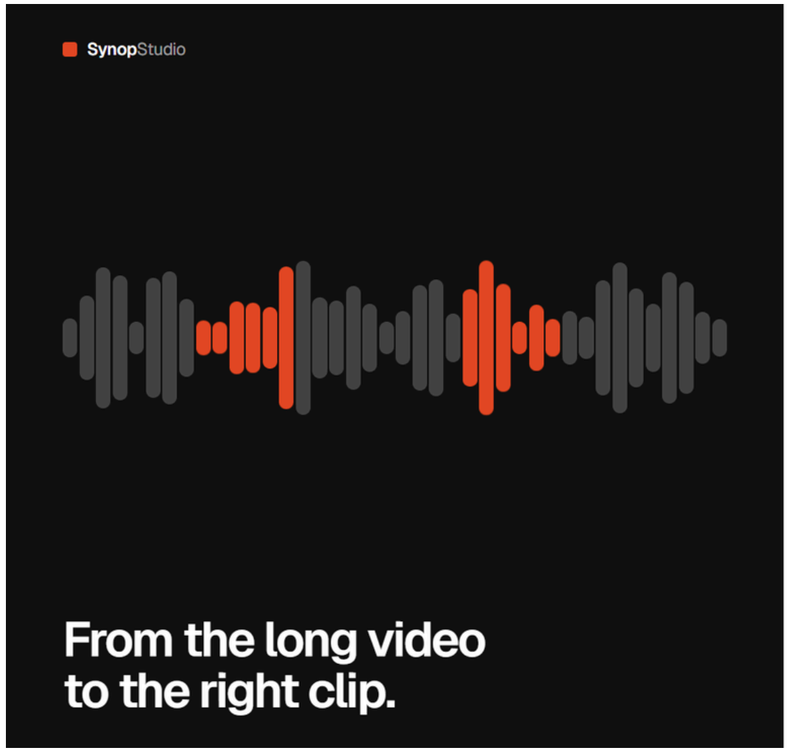
User profiles

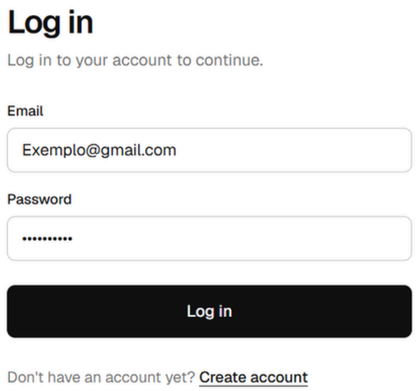
6 2.1.2 · Who we design for

Profile	Who	Main barrier	WCAG	Test question
Blind users	Viewer	Icon-only buttons, visual timeline	4.1.2	Screen reader only?
Low vision	Viewer	Small text, thin borders	1.4.3	Readable at 200% zoom?
Color blindness	Viewer	Selection marked only in orange	1.4.1	Clear without color?
Motor impairments	Creator	Drag-only timeline	2.5.7	Keyboard only?
Deafness	Creator	Captions only in the editor	1.2.2	Works without sound?
Cognitive / low literacy	Creator	Editing jargon, long forms	3.3.2	Clear on first try?
How we test	All	Screen reader · zoom · keyboard · mute	2.2 AA	One question per screen

Accessibility matrix - Log In Page

3 **2 Medium issues and 1 Low issue**





1 Low Visual Borders (Medium)

borders are very light grey on white (1.4.11) →contrast of at least 3:1

2 Decorative Panel (Low)

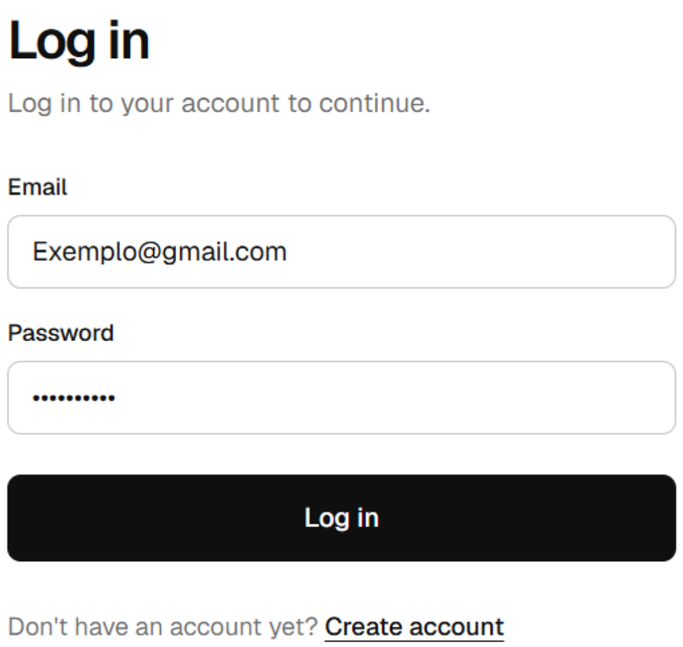
maybe exposed to screen readers(1.1.1) → aria-hidden

3 Split Screen Layout (Medium)

Left panel keeps half of the screen at zoom 200%(1.4.10) → hide or stack the decorative panel

Accessibility matrix - Log In Page

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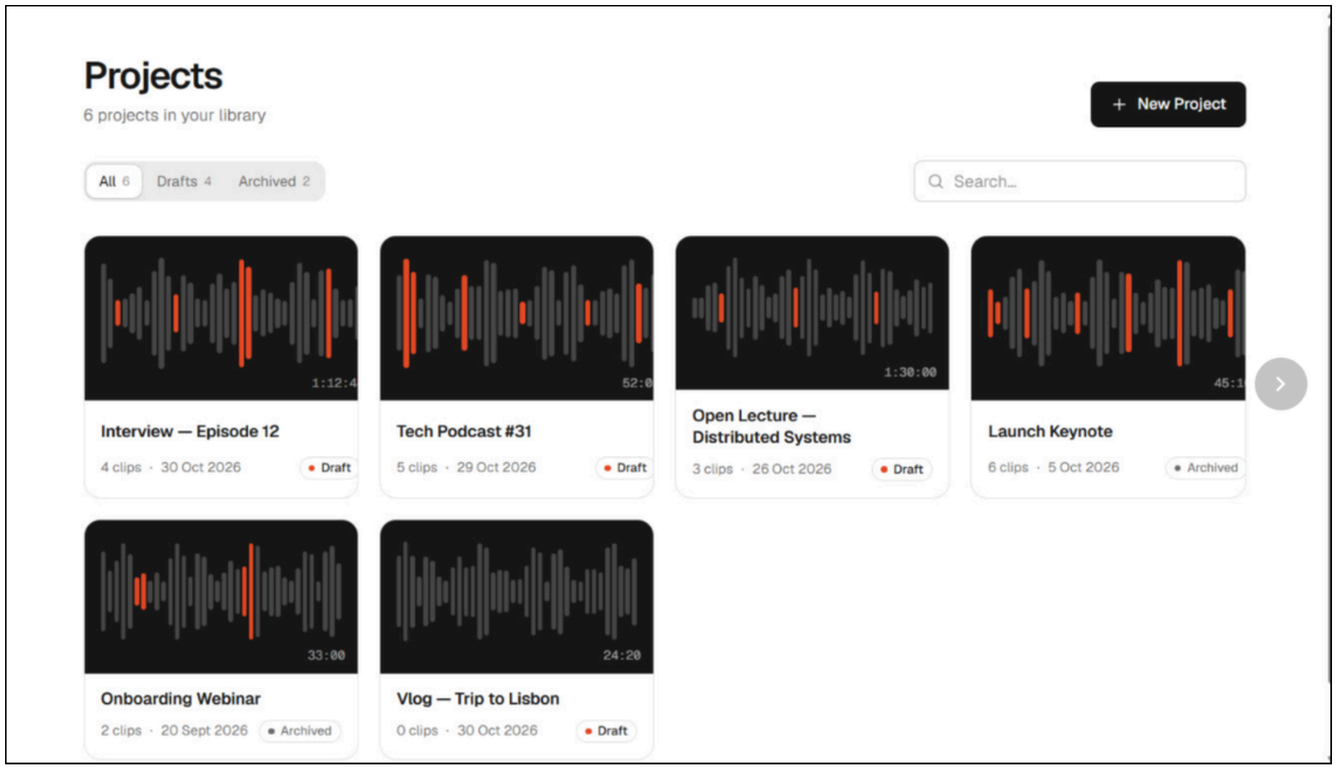
Left panel keeps half of the screen at zoom 200%(1.4.10) → hide or stack the decorative panel

Accessibility matrix - Log in Page

Element	User Profile	Problem	WCAG	Severity	Correction
Email and Password fields	Low Vision	Input field borders are very light grey on white (estimated ~1.5:1), making the field boundaries hard to perceive	1.4.11 Non-text Contrast	Medium	Darken field borders to a contrast of at least 3:1 against the background
Decorative waveform panel	Blind Users	The waveform illustration is purely decorative but may be exposed to screen readers as a series of graphic elements, adding noise before the form	1.1.1 Non-text Conten	Low	Mark the illustration as decorative
Split-Screen layout	Low Vision	At 200% zoom, the decorative left panel keeps half of the screen, which may squeeze the form or force horizontal scrolling	1.4.10 Reflow	Medium	At narrow widths or high zoom, hide or stack the decorative panel so the form uses the full width

Accessibility matrix - Project List Page

3 **3 high-severity issues**



1 Project cards (High)

Card is clickable but is not a link → One link per card, named by the project name

2 Search field (High)

Only a placeholder, no name → Give the field a spoken name

3 Filters and project count (High)

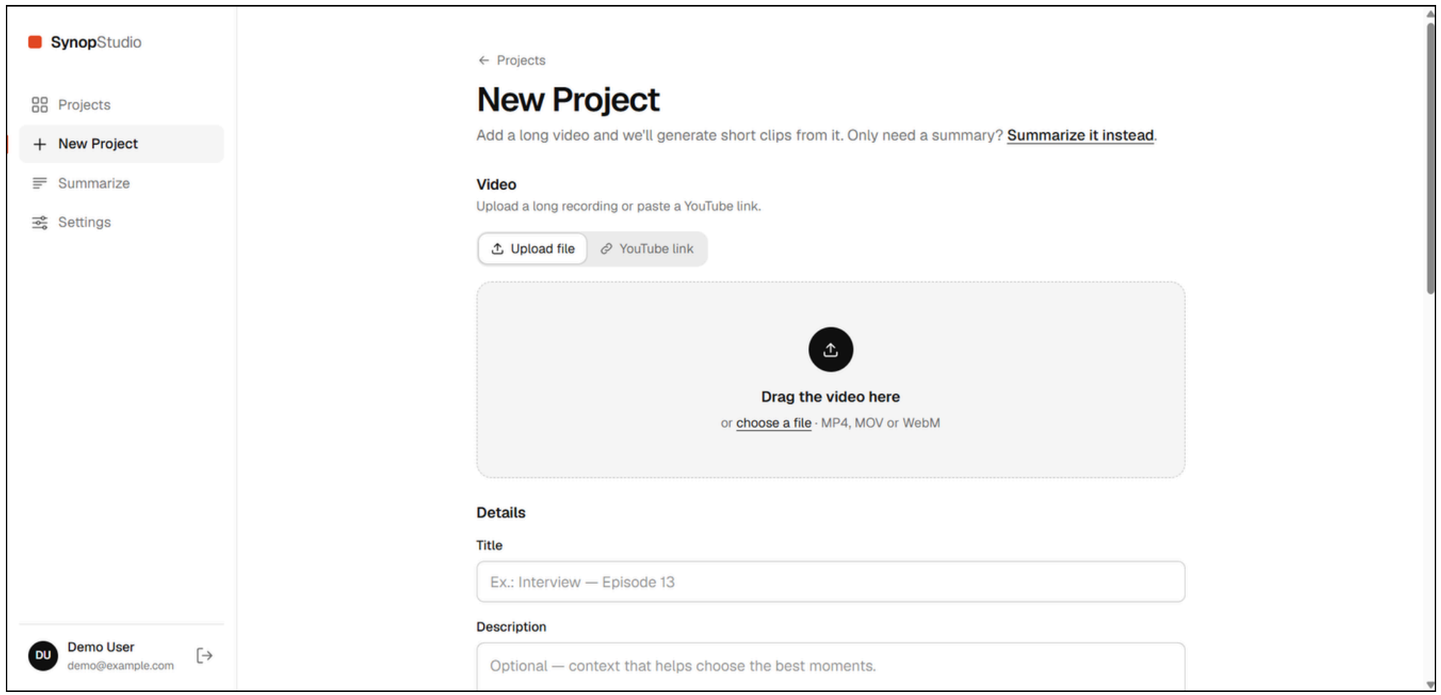
State and count not exposed → Say which filter is selected and the number of projects

Accessibility matrix - Project List Page

Element	User Profile	Problem	WCAG	Severity	Correction
Project cards	Blind Users, Low Vision, Motor Impairments	The whole card is clickable but is not a link, so it is read as loose text	4.1.2 (Name, Role, Value) 2.4.4 (Link purpose (In Context))	High	One link per card, named by the project title
Search field	Blind Users, Low Vision	Only the placeholder “Search…” and an icon, so it has no reliable name	4.1.2 (Name, Role, Value)	High	Give the field a spoken name: “Search projects”
Filters and count	Blind Users, Low Vision	Active filter shown only by its background, the new project count is not announced	4.1.2 (Name, Role, Value) 4.1.3 (Status Messages)	High	Say which filter is selected and read out the new number of projects

Accessibility matrix - Create Project Page

3 **2 Medium issues and 1 High issue**



1 Video Upload State (Medium)

large uploads lack progress and completion announcements (4.1.3) → progress bar, live regions and “Cancel Upload” button

2 Format and Style Focus (High)

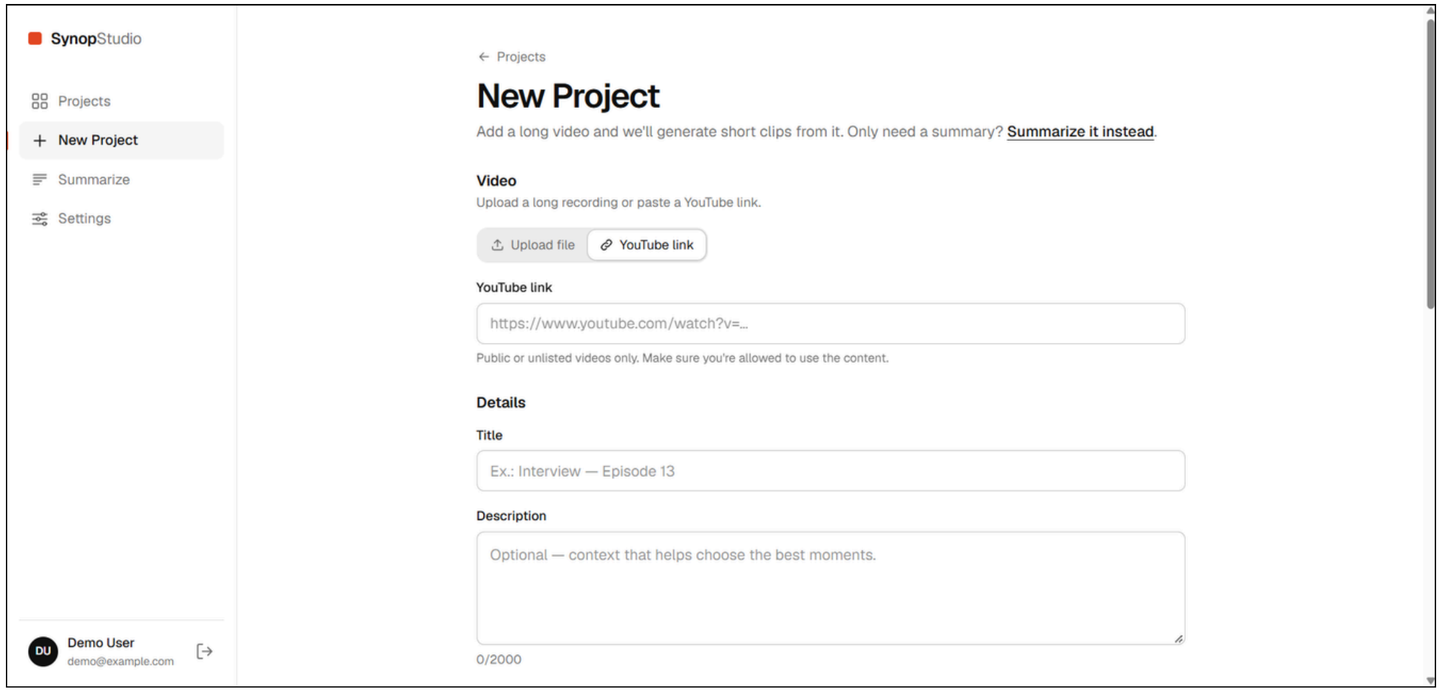
orange focus ring is the same as selection (2.4.7) → distinct focus ring and grouping with fieldset/legend

3 Form Fields (Medium)

required fields and size limits unmarked (3.3.2) → write “(required)” in text and size limits hints with aria-describedby.

Accessibility matrix - Create Project Page

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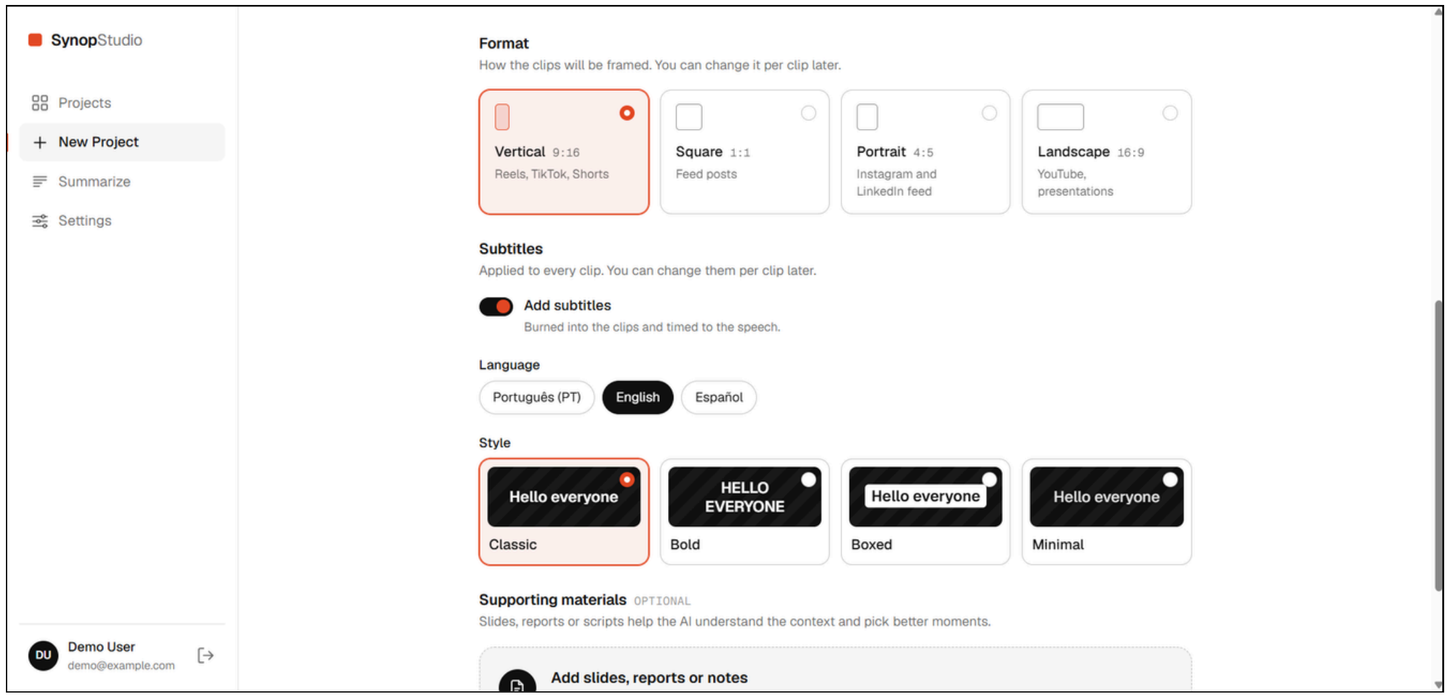
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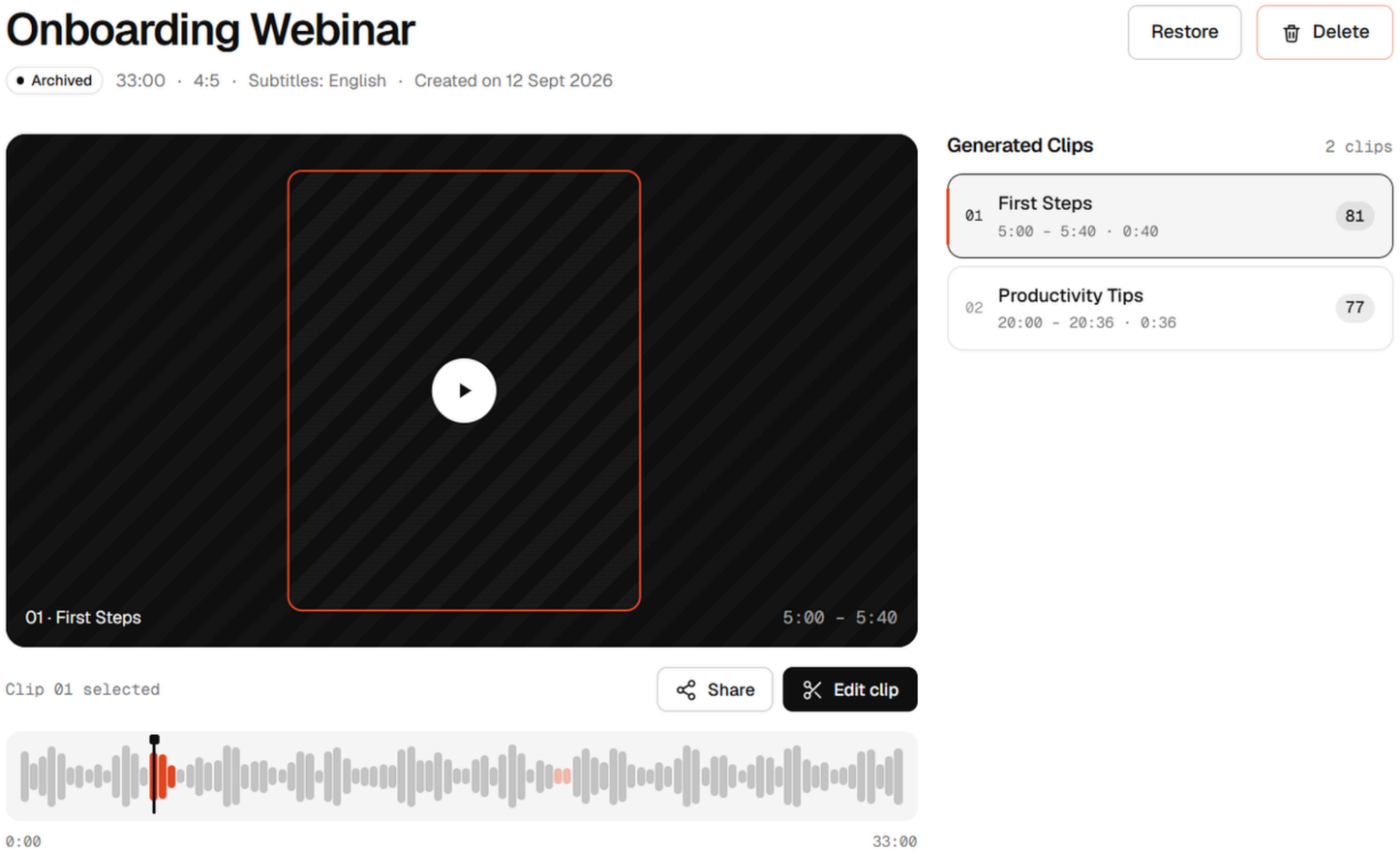
required fields and size limits unmarked (3.3.2) → write “(required)” in text and size limits hints with aria-describedby.

Accessibility matrix - Create Project Page

Element	User Profile	Problem	WCAG	Severity	Correction
Video Upload	Blind users; Cognitive/low digital literacy	Progress and completion status of long uploads are not announced to screen readers	4.1.3 Status Messages	Medium	Progress bar, live regions for status updates and a “Cancel upload” button
Format, Style, Language and Source toggles	Motor impairments	The focus ring shares the same orange color as the selected state, making the focus invisible on active cards	2.4.7 Focus Visible	High	Distinct focus ring and native grouping using fieldset/legend
Title, Video, Link, Supporting materials	Blind users; Cognitive/low digital literacy	Required fields are unmarked, size limits are missing and rules rely solely on placeholders	3.3.2 Labels or Instructions	Medium	Explicit “(required)” text labels, limit hints, and aria-describedby associations.

Accessibility matrix - Clip Proposals Page

3 2 High issues and 1 Medium issue



1 No Captions (High)

The player has no captions or captions button
→ captions on by default, with a toggle

2 No Transcript (High)

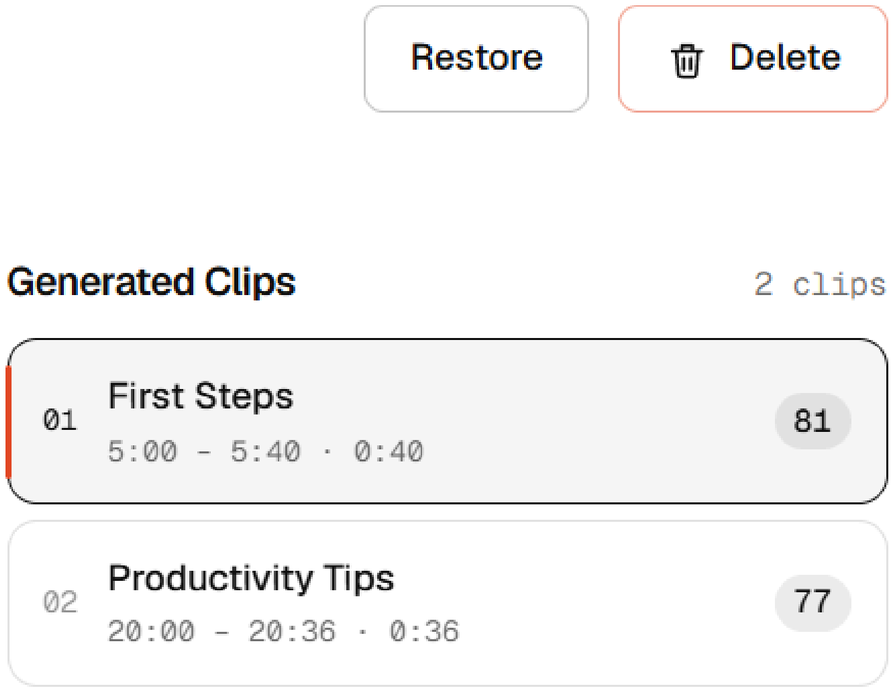
No synchronized transcript next to the player
→ transcript panel with speaker names

3 Reframing Shown Only by Color (Medium)

The 4:5 crop is marked only by an orange outline
→ text label and patterned outline

Accessibility matrix - Clip Proposals Page

3 **2 High issues and 1 Medium issue**



1 No Captions (High)

The player has no captions or captions button
→ captions on by default, with a toggle

2 No Transcript (High)

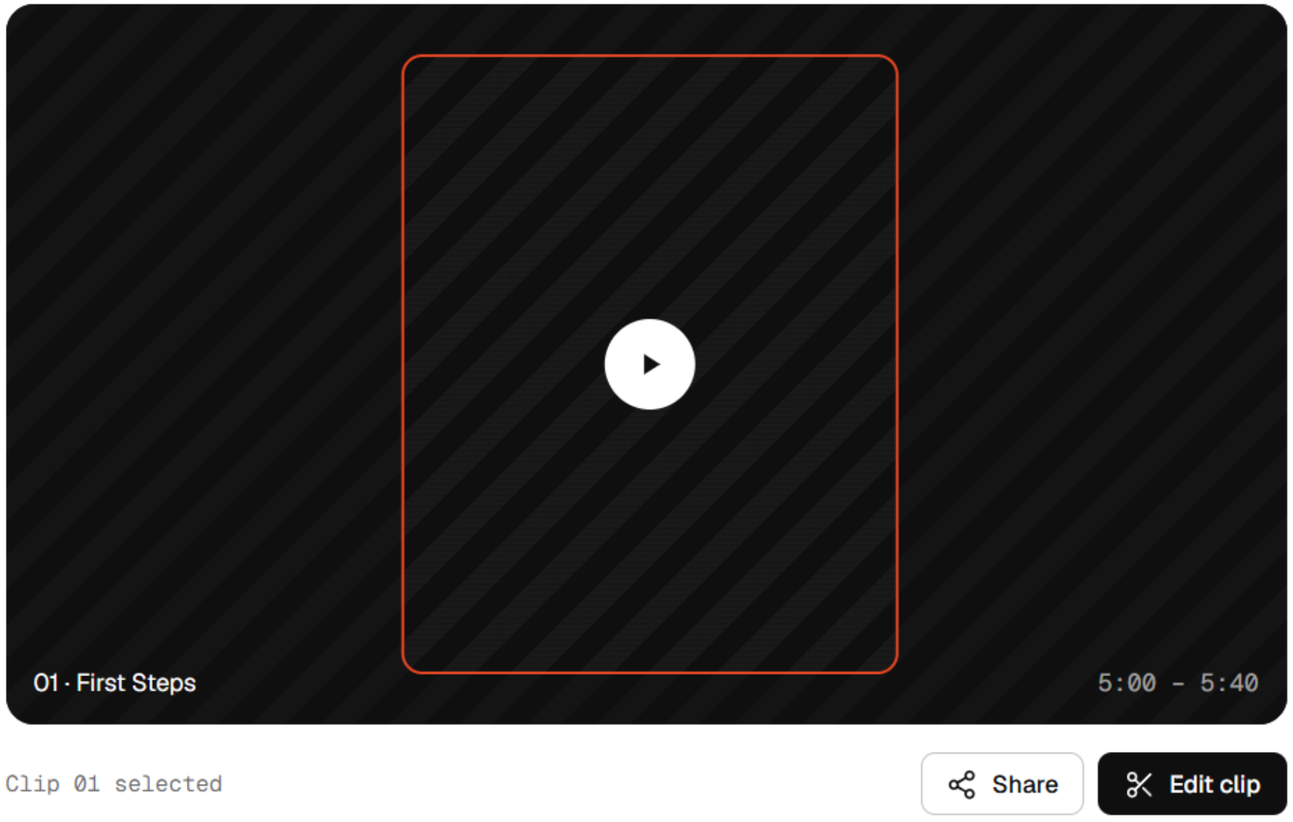
No synchronized transcript next to the player
→ transcript panel with speaker names

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Accessibility matrix - Clip Proposals Page

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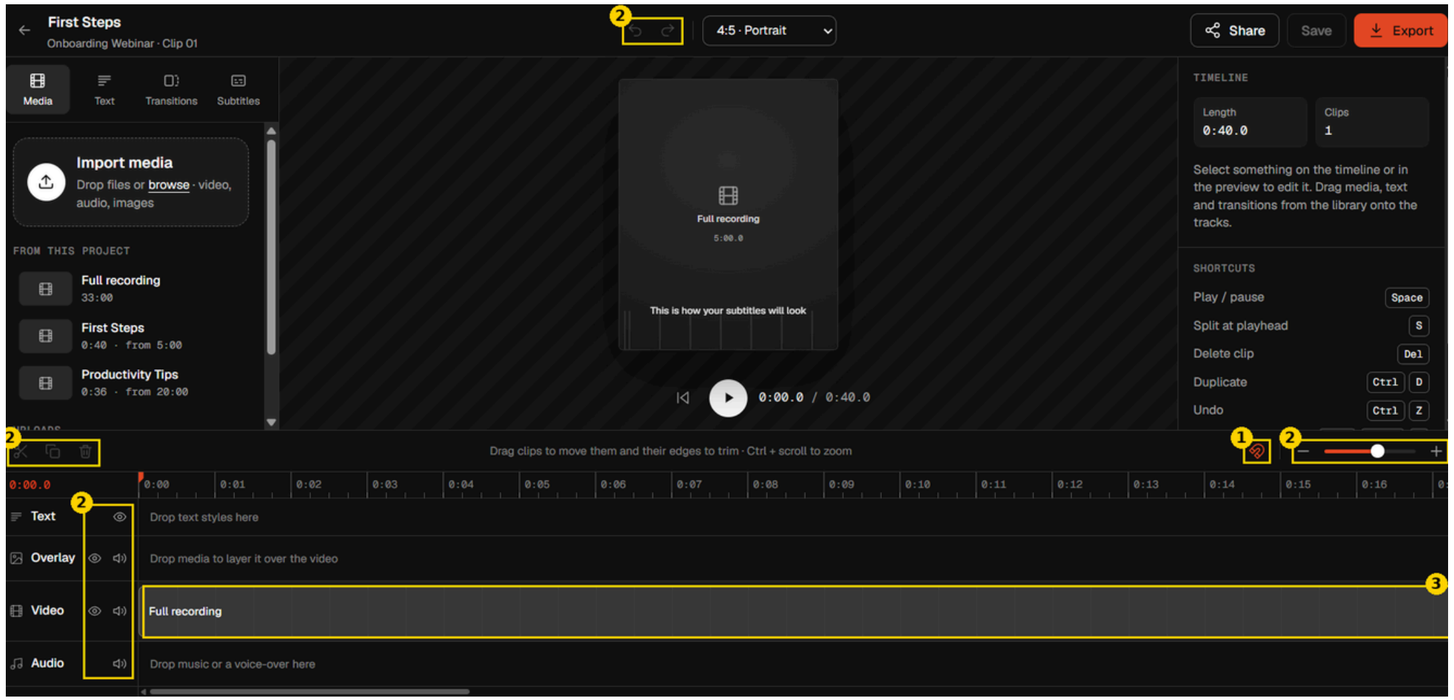
The 4:5 crop is marked only by an orange outline
→ text label and patterned outline

Accessibility matrix - Clip Proposals Page

Element	User Profile	Problem	WCAG	Severity	Correction
Player (captions)	Deafness	No captions or captions button. A deaf creator cannot check what is said before approving the clip	1.2.2 Captions (Prerecorded)	High	Show captions on by default, with a button to turn them on and off
Player (transcript)	Deafness, Blind Users	No transcript next to the player, so the recording has no text alternative and speakers are not identified	1.2.3 Audio Description or Media Alternative; 1.2.2 Captions (Prerecorded)	High	Synced transcript with speaker names, synchronized with the player and highlighting the selected clip's interval
Player (reframing area)	Blind Users, Color Blindness	The 4:5 crop is shown only by an orange outline	1.1.1 Non-text Content; 1.4.1 Use of Color	Medium	Text label and patterned outline

Accessibility matrix - Clip Editor

3 Three high-severity issues



1 Snapping toggle

On/off shown only by color (1.4.1) → filled state + label

2 Icon-only buttons

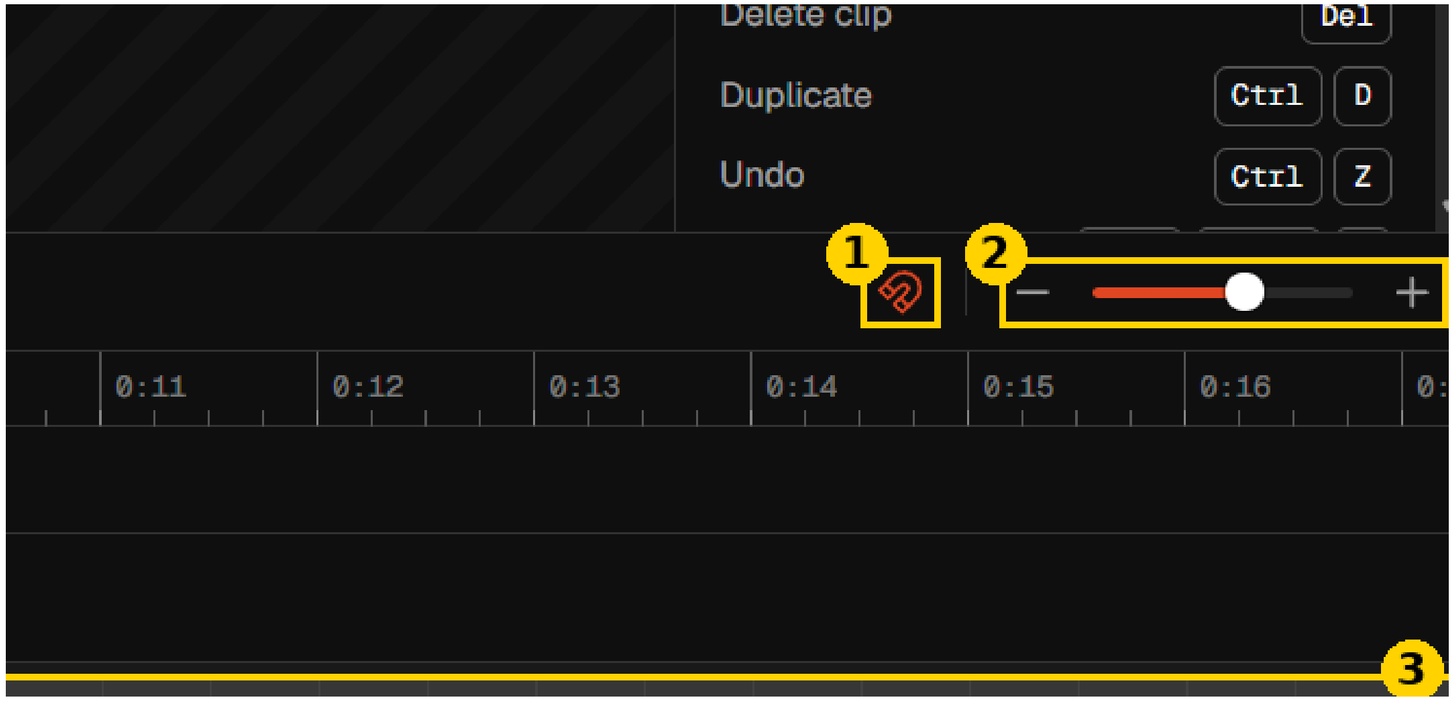
No accessible name (4.1.2) → aria-label + tooltip

3 Clips on the timeline

Drag only (2.5.7) → arrow keys + typed times

Accessibility matrix - Clip Editor

3 Three high-severity issues



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On/off shown only by color (1.4.1) → filled state + label

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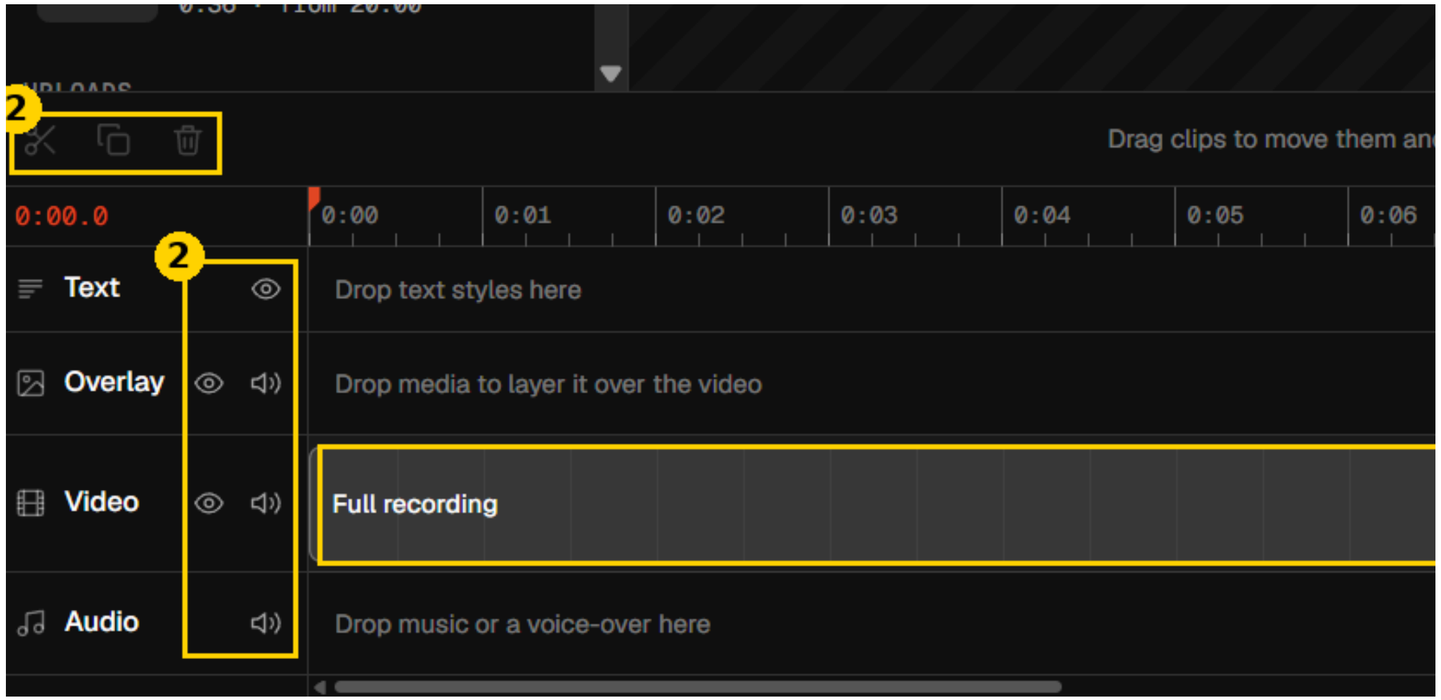
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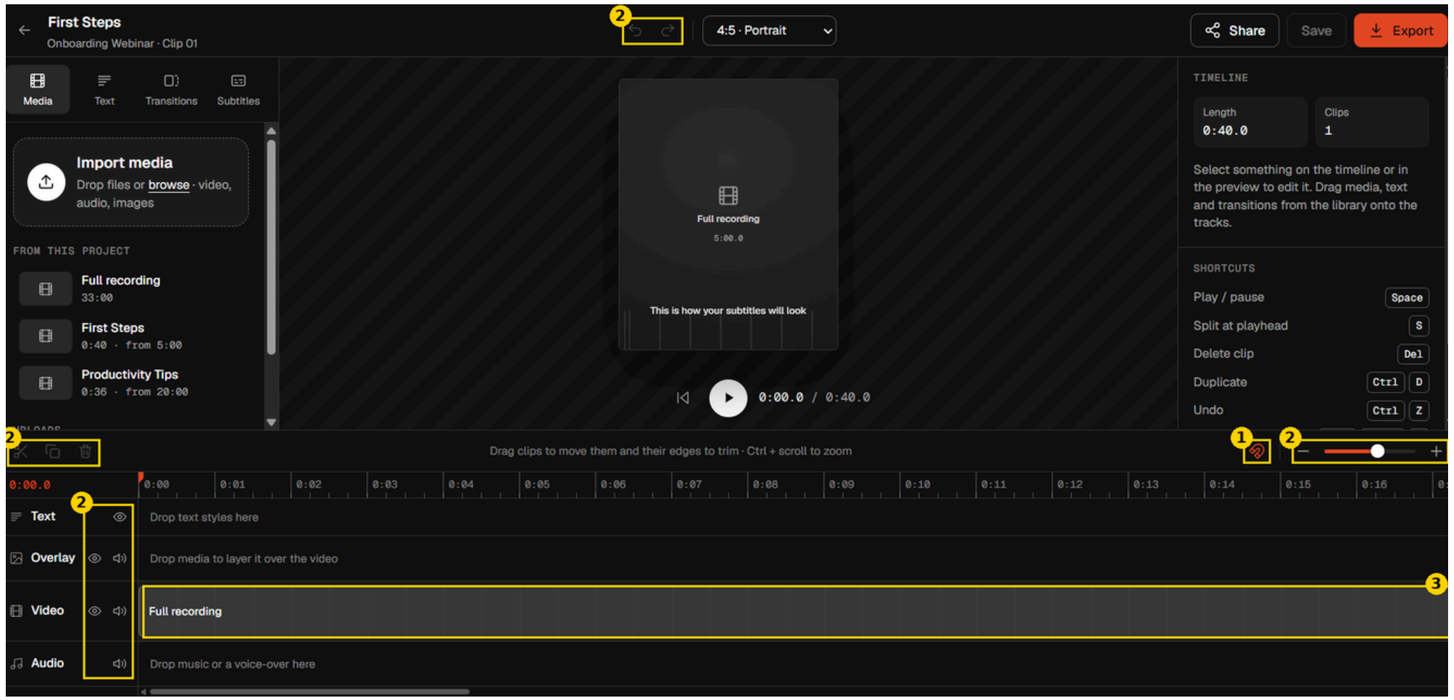
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Accessibility matrix - Clip Editor

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No accessible name (4.1.2) → aria-label + tooltip

3 Clips on the timeline

Drag only (2.5.7) → arrow keys + typed times

Accessibility matrix - Clip Editor

Element	User Profile	Problem	WCAG	Severity	Correction
Snapping toggle (magnet)	Color Blindness, Blind Users	On/off state shown only by the icon turning orange, with no label or tooltip	1.4.1 Use of Color; 4.1.2 Name, Role, Value	High	Filled background when on, "Snapping on/off" tooltip and pressed state in code
Icon-only buttons (toolbar, tracks, zoom)	Blind Users	Buttons show only an icon, so a screen reader has no name to announce	4.1.2 Name, Role, Value	High	Accessible name on each button ("Split clip", "Hide track") and a matching tooltip
Clips on the timeline	Motor Impairments	Clips are moved and trimmed only by dragging	2.5.7 Dragging Movements; 2.1.1 Keyboard	High	Arrow keys move the selected clip; start and end times can be typed

MS3 **14/10/2026 - 3/11/2026**	**Module: UI/UX, Accessibility & Usability** T3.1 - Accessibility Standards and User Profiles Framework – Lucas and Daniel T3.2 - Screen Reader and Semantics Accessibility Audit - Murilo T3.3 - Color Contrast, Daltonism, and Visual Scaling Audit - Daniel T3.4 - Keyboard Navigation, Focus, and Form Interaction Audit - Tomás T3.5 - Video Captions and Media Accessibility Verification - David T3.6 - Screen-by-Screen Accessibility Matrix and Remediation Report - Everyone **Module: Legal and Regulatory Requirements** T3.7 - GDPR and Data Protection Compliance Specification - Lucas T3.8 - Software Dependencies and Open-Source Licensing Audit - Murilo	Ensure legal and regulatory compliance, conduct screen-by-screen accessibility audits, while building the core backend and AI pipeline.	D3.1 - Legal Compliance, Usability and Accessibility Evaluation Report D3.2 - Core Backend Services and Data Pipelines D3.3 - AI Ingestion and Acoustic Pipeline Prototype D3.4 - MS3 PowerPoint

	T3.9 - Sector-Specific Regulations and Standards Analysis - David T3.10 - Terms of Service and Liability Disclaimers Definition - Tomás		
	Module: Backend and Infrastructure T3.11 - Authentication and User Accounts - Daniel T3.12 - Video Upload and Project Management – David e Lucas T3.13 - Async Job Processing and Status - Tomás e Murilo		
	Module: AI Pipeline T3.14 - Transcription & Speaker Identification – David e Murilo T3.15 - Clip Proposal Generation - Everyone		
	Module: Project Management T3.16 - MS3 Presentation - Everyone		
MS4 **4/11/2026 - 15/12/2026**	**Module: Video Editor and Export** T4.1 - Clip Proposals Page and Traceability – Daniel T4.2 - Text-based Editor - Lucas T4.3 - Rendering, Captions and Export – Lucas e Daniel	Deliver a working, tested and documented MVP, with AI model evaluation, presented to supervisors with peer evaluation.	D4.1 - Minimum Viable Product (MVP) Software Release (v1.0)
	Module: AI Pipeline		

	T4.4 - AI Metadata Generation – Murilo **Module: Backend and Infrastructure** T4.5 - Privacy, Audit Log and Data Deletion – Murilo e David **Module: UI/UX, Accessibility and Usability** T4.6 - Accessibility and Usability Fixes – David e Tomás **Module: Testing and Validation** T4.7 - Evaluation Dataset – David and Tomás T4.8 - AI Model Evaluation - Everyone T4.9 - Performance and Security Tests - Daniel T4.10 - Stabilization and Bug Fixing - Everyone **Module: Project Management** T4.11 - MVP Scope Freeze - Everyone T4.12 - Checkpoint Demo - Daniel T4.13 - Documentation and Final Report - Everyone T4.14 - Demo Preparation - Everyone T4.15 - MS4 Presentation - Everyone		D4.2 - AI Models Validation and Benchmark Report D4.3 - System Quality, Security and Scalability Report D4.4 - Agile Project Management D4.5 - Technical Documentation D4.6 - MS4 Live Demo and PowerPoint

Thank you.

https://pei-synopstudio.github.io/docs/

